

Electromagnetism

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1 Lessons

1.1 Electromagnetism: charges, fields, and the link between them

A battery, a magnet, and a beam of light look like three different topics. They are three faces of one interaction. Electric charges act on each other, moving charges produce magnetism, and a changing magnetic field produces an electric field and vice versa—which is how generators, transformers, and radio work. This lesson covers the electromagnetism shared by FY6 (charge, electric field, potential, capacitors, simple DC circuits) and FY7 (magnetic field, the force on a moving charge, induction). The optics half of FY7 is left for another lesson.

The organising idea of the whole topic is the *field*. A charge does not pull on a distant charge across empty space; it creates a field at every point of space, and that field acts on whatever charge happens to sit there. In code terms a field is a function from positions to force-per-unit-charge vectors. The analogy is useful because superposition in physics is exactly linearity in code: compute the contribution of each charge and add. Where the analogy breaks down is discussed at the end.

A route through the topic:

1. charge and Coulomb’s law;
2. the electric field;
3. electric potential and energy;

4. conductors and capacitors;
5. current and simple DC circuits;
6. the magnetic field and the force on a moving charge;
7. magnetic flux and induction;
8. the unified picture: Maxwell's equations and electromagnetic waves.

Constants used throughout: $k = 9.0 \times 10^9 \text{ N m}^2/\text{C}^2$ (often written $k = 1/(4\pi\epsilon_0)$, with $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$), $e = 1.60 \times 10^{-19} \text{ C}$, $m_e = 9.11 \times 10^{-31} \text{ kg}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$, $c = 3.0 \times 10^8 \text{ m/s}$.

1.1.1 Charge and Coulomb's law

Definition 1.1 (Coulomb's law). Two point charges q_1 and q_2 a distance r apart attract or repel each other along the line joining them, with force magnitude

$$|F| = k \frac{|q_1 q_2|}{r^2}.$$

Like signs repel, opposite signs attract. The forces on the two charges are an action–reaction pair: equal magnitude, opposite direction. If more than two charges are present, the total force on any one of them is the vector sum of the pairwise forces (*superposition*).

Charge is quantised in nature, $q = ne$ with $n \in \mathbb{Z}$, but at this level we treat q as a continuous quantity: one coulomb is an enormous number of elementary charges, so the discreteness only matters in single-particle physics. The inverse-square form is the same shape as Newton's gravity, but the analogy breaks down immediately in one respect: gravity has only one sign, so masses only attract, while charges of both signs exist and fields can cancel.

Example 1.2 (Force between two charges). Charges $q_1 = 2.0 \text{ } \mu\text{C}$ and $q_2 = -3.0 \text{ } \mu\text{C}$ sit 30 cm apart. Then

$$|F| = 9.0 \times 10^9 \cdot \frac{(2.0 \times 10^{-6})(3.0 \times 10^{-6})}{(0.30)^2} = 0.60 \text{ N}.$$

The signs are opposite, so the force pulls the charges together: 0.60 N on each, toward the other one. Note that the force does not depend on what, if anything, lies between the charges.

1.1.2 The electric field

Definition 1.3 (Electric field). The electric field at a point is the force experienced by a small positive test charge q_0 placed there, per unit charge:

$$\vec{E} = \frac{\vec{F}}{q_0}, \quad \text{unit N/C} = \text{V/m}.$$

The field of a point charge q at distance r points radially away from (or toward, for $q < 0$) the charge and has magnitude

$$E = k \frac{|q|}{r^2}.$$

Fields of several charges add as vectors (superposition).

Example 1.4 (Two equal charges). Two identical charges $q = 1.0 \text{ nC}$ sit at $(\pm 5.0 \text{ cm}, 0)$. At the midpoint the fields cancel exactly and $E = 0$, which is a useful warning: zero field does not mean no charges nearby. At the point $P = (0, 10 \text{ cm})$ each charge is $r = \sqrt{(0.050)^2 + (0.10)^2} = 0.112 \text{ m}$ away, so each contributes

$$E_1 = 9.0 \times 10^9 \cdot \frac{1.0 \times 10^{-9}}{0.0125} = 720 \text{ N/C}.$$

The horizontal components cancel by symmetry; the vertical ones add with $\cos \theta = 0.10/0.112 = 0.894$:

$$E = 2 \cdot 720 \cdot 0.894 \approx 1.3 \text{ kN/C},$$

pointing away from the midpoint along $+y$.

Between two large parallel plates with voltage U and separation d the field is uniform (constant in magnitude and direction):

$$E = \frac{U}{d}. \quad (1)$$

For example, $U = 200 \text{ V}$ across 2.0 cm gives $E = 1.0 \times 10^4 \text{ V/m}$.

Theorem 1.5 (Static charge lives on the surface of a conductor). *In electrostatic equilibrium the electric field inside a solid conductor is zero, any excess charge resides on the outer surface, and the whole conductor is at one and the same potential.*

Proof sketch. If $\vec{E} \neq 0$ inside the metal, a free electron would feel $\vec{F} = q\vec{E} \neq 0$ and move—so the situation would not be static. Hence $\vec{E} = 0$ inside. If some excess charge sat in the interior, field lines would run from it (or to it) through the metal, giving $\vec{E} \neq 0$ nearby; the charges therefore migrate to the surface until no field remains inside. Finally, if two points of the conductor had different potentials, a charge could be moved between them with the field doing net work, meaning a field component along the metal—again motion. So the potential is constant throughout. $\square$

This is why a car is a safe place in a lightning storm (a metal shell keeps its interior field-free) and why sensitive electronics live in metal enclosures.

1.1.3 Potential and energy

Definition 1.6 (Potential, potential energy, voltage). Moving a charge q in a field changes its potential energy by

$$\Delta W_{\text{pot}} = q \Delta V, \quad \Delta V = V_B - V_A,$$

where V is the *electric potential*, unit volt ($\text{V} = \text{J/C}$). The voltage between A and B is the work the field does per unit charge moving from A to B :

$$U_{AB} = V_A - V_B = \frac{W_{AB}}{q}.$$

In a uniform field of strength E , the potential difference across a distance d measured along the field lines is

$$U = Ed,$$

and the potential always *decreases* in the direction the field points.

Example 1.7 (Electron accelerated through 5.0 kV). An electron released from rest and pulled through $U = 5.0 \text{ kV}$ loses potential energy

$$\Delta W_{\text{pot}} = eU = (1.60 \times 10^{-19})(5.0 \times 10^3) = 8.0 \times 10^{-16} \text{ J},$$

which becomes kinetic energy $\frac{1}{2}m_e v^2$. Solving,

$$v = \sqrt{\frac{2eU}{m_e}} = 4.2 \times 10^7 \text{ m/s}.$$

This is about 14% of the speed of light, so the non-relativistic formula is already drifting by a percent or so—fine for high school, but a real accelerator needs Einstein’s version.

Remark (Three quantities that are easy to mix up). $\vec{E}$ is force per unit charge (V/m), V is potential energy per unit charge (V), and $W_{\text{pot}} = qV$ is energy (J). A handy unit appears here: one *electron-volt* is the energy an electron gains through one volt, $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$. Also mind the signs: a positive charge “wants to fall” toward lower potential (it speeds up going downhill), a negative charge speeds up going *up* in potential. Equations stay correct if you let q carry its sign and never take absolute values prematurely.

1.1.4 Conductors and capacitors

Definition 1.8 (Capacitance). A capacitor stores charge Q separated by a voltage U , with

$$C = \frac{Q}{U}, \quad \text{unit farad } \text{F} = \text{C/V}.$$

For a parallel-plate capacitor with plate area A , separation d , and dielectric (insulator) of relative permittivity ϵ_r ,

$$C = \epsilon_0 \epsilon_r \frac{A}{d}. \quad (2)$$

C is a property of the geometry and the insulator, not of how much charge you put on—it answers the question “how much voltage do I pay per unit of stored charge?”

Proposition 1.9 (Energy stored in a capacitor). *A capacitor charged to charge Q and voltage $U = Q/C$ stores energy*

$$W = \frac{Q^2}{2C} = \frac{1}{2}CU^2.$$

Proof. Charge the capacitor in infinitesimal slices. When it already holds charge q , its voltage is $u = q/C$, so moving the next slice dq against that voltage takes work $dW = u dq = (q/C) dq$. Summing over the whole charge:

$$W = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C} = \frac{1}{2}CU^2. \quad \square$$

Example 1.10 (A parallel-plate capacitor). Plates of area $A = 100 \text{ cm}^2 = 1.0 \times 10^{-2} \text{ m}^2$, separated by $d = 1.0 \text{ mm}$ of plastic with $\epsilon_r = 3.0$:

$$C = (8.85 \times 10^{-12})(3.0) \frac{1.0 \times 10^{-2}}{1.0 \times 10^{-3}} = 2.7 \times 10^{-10} \text{ F} = 0.27 \text{ nF}.$$

At $U = 12 \text{ V}$ this holds $Q = CU = 3.2 \text{ nC}$ and energy

$$W = \frac{1}{2}CU^2 = 1.9 \times 10^{-8} \text{ J}.$$

Tiny on the scale of a hot drink, large on the scale of a camera flash, which dumps exactly this kind of energy in a millisecond.

Remark (Breakdown and safety). If the field exceeds the *breakdown strength* of the insulator (roughly 3 MV/m for air), the dielectric ionises and conducts—a spark. This is why capacitors and cables have voltage ratings, and why FY6 emphasises fuses, protective grounding, and enclosure classes: the physics of breakdown is what safety regulations are written around. A capacitor can hold a lethal charge long after power is removed.

1.1.5 Current and simple DC circuits

Definition 1.11 (Current, resistance, power). Electric current is charge in motion,

$$I = \frac{\Delta Q}{\Delta t}, \quad \text{unit ampere } A = C/s.$$

A resistor obeys *Ohm's law*,

$$U = RI, \quad \text{unit ohm } \Omega = V/A,$$

and a component with voltage U across it and current I through it converts electrical energy at power

$$P = UI = I^2 R = \frac{U^2}{R}, \quad \text{unit watt.}$$

By convention current flows from $+$ to $-$ around the circuit, while in metal wires the electrons actually drift the other way. Only the sign convention matters for the calculations.

Proposition 1.12 (Series, parallel, and the voltage divider). *Resistors in series add, $R = R_1 + R_2 + \dots$, and resistors in parallel combine as*

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

In particular, two resistors in series across a source U share the voltage in proportion to their resistances:

$$U_2 = U \frac{R_2}{R_1 + R_2}. \quad (3)$$

Proof. In series the same current I flows through both, and the voltages add: $U = IR_1 + IR_2 = I(R_1 + R_2)$, so $R = R_1 + R_2$. In parallel both see the same voltage U , and the currents add:

$$I = \frac{U}{R_1} + \frac{U}{R_2} = U \left(\frac{1}{R_1} + \frac{1}{R_2} \right),$$

giving the parallel formula. The divider formula is $U_2 = IR_2$ with $I = U/(R_1 + R_2)$. $\square$

Example 1.13 (A small network). A 12 V source drives $R_1 = 100 \Omega$ in series with the parallel pair $R_2 = 200 \Omega$, $R_3 = 300 \Omega$. The pair combines to

$$R_{23} = \frac{R_2 R_3}{R_2 + R_3} = 120 \Omega,$$

so the source sees $R = 220 \Omega$ and the main current is $I = 12/220 = 55 \text{ mA}$. The power delivered is $P = UI = 0.65 \text{ W}$. By the divider formula the parallel pair takes $U_{23} = 12 \cdot 120/220 = 6.5 \text{ V}$, so $I_2 = 33 \text{ mA}$ and $I_3 = 22 \text{ mA}$ (they sum to 55 mA, a good self-check).

1.1.6 The magnetic field and the force on a moving charge

Definition 1.14 (Magnetic field and the Lorentz force). A magnetic field exerts on a charge q moving with velocity $\vec{v}$ the force

$$\vec{F} = q \vec{v} \times \vec{B}, \quad |F| = |q| v B \sin \alpha, \quad (4)$$

where α is the angle between $\vec{v}$ and $\vec{B}$. The unit of B is the tesla, $T = N/(A \cdot m)$. The direction is given by the right-hand rule: point fingers along $\vec{v}$, curl them toward $\vec{B}$; $\vec{F}$ is the thumb, reversed in sign for a negative charge.

A straight wire of length L carrying current I at angle α to the field feels $|F| = BIL \sin \alpha$, the wire form of the same law ($\vec{v}$ replaced by the drift of the charge carriers).

Two features of eq. (4) deserve emphasis. First, a charge at rest feels nothing—magnetism is a relativistic aspect of the same interaction, which is why a current balance measures electric and magnetic forces against each other. Second, since $\vec{F} \perp \vec{v}$ always, the magnetic field never changes the speed of a particle, only its direction. This is the content of exercise 2.6.

Example 1.15 (Proton in a uniform field). A proton enters $B = 0.40$ T at $v = 3.0 \times 10^5$ m/s, perpendicular to the field. The force $|F| = qvB$ is always perpendicular to the motion, so it is a centripetal force and the path is a circle:

$$qvB = \frac{m_p v^2}{r} \implies r = \frac{m_p v}{qB} = \frac{(1.67 \times 10^{-27})(3.0 \times 10^5)}{(1.60 \times 10^{-19})(0.40)} = 7.8 \text{ mm}.$$

The period of one lap is

$$T = \frac{2\pi r}{v} = \frac{2\pi m_p}{qB} = 0.16 \text{ } \mu\text{s},$$

and remarkably it does not depend on v (as long as $v \ll c$): faster particles just travel bigger circles in the same time. This is the operating principle of the cyclotron.

1.1.7 Magnetic flux and induction

Definition 1.16 (Magnetic flux). The magnetic flux through a flat coil of area A is the amount of field passing through it,

$$\Phi = BA \cos \alpha,$$

where α is the angle between $\vec{B}$ and the normal (perpendicular) of the coil. The unit is the weber, $\text{Wb} = \text{T m}^2$. For a coil of N turns one usually counts the total flux $N\Phi$.

Definition 1.17 (Faraday's law and Lenz's law). A changing magnetic flux induces a voltage,

$$U_{\text{ind}} = -N \frac{\Delta \Phi}{\Delta t}, \quad (5)$$

with the sign fixed by *Lenz's law*: the induced current creates a field that opposes the change in flux. If the flux increases into the coil, the induced current pushes a field out of it; the induced current always fights whatever is changing.

Lenz's law is not bookkeeping pedantry—it is conservation of energy in disguise. If the induced current *helped* the change, the flux would grow, inducing a larger current, growing the flux further: free energy from nothing. Opposing the change is what makes you do mechanical work to turn a generator.

Example 1.18 (A coil in a growing field). A coil of $N = 200$ turns, area 4.0 cm^2 , sits perpendicular to a field which grows from 0 to $B = 0.10$ T in $\Delta t = 20$ ms. The flux grows to $\Phi = BA = 4.0 \times 10^{-4}$ Wb, so

$$U_{\text{ind}} = N \frac{\Delta \Phi}{\Delta t} = 200 \cdot \frac{4.0 \times 10^{-4}}{2.0 \times 10^{-2}} = 4.0 \text{ V}.$$

The direction: the field into the coil is growing, so the induced current drives a field out of the coil (right-hand rule: curl your fingers in the current direction and the thumb points out).

Two classic applications close FY7's electromagnetism. If the coil rotates at angular frequency ω in a fixed field, the flux oscillates and eq. (5) gives an alternating voltage

$$U_{\text{ind}}(t) = NBA\omega \sin(\omega t)$$

—the AC generator. And if a rod of length L slides along rails at speed v perpendicular to a field, the flux swept per second is BLv , so the induced voltage is

$$U_{\text{ind}} = BLv.$$

1.1.8 One interaction: Maxwell's equations and electromagnetic waves

Everything above condenses into four statements—Maxwell's equations—which in words say:

- (i) electric charges create electric fields (field lines start and end on charges);
- (ii) there are no magnetic charges: field lines never start or end, they close on themselves;
- (iii) a changing magnetic field creates an electric field (theorem 1.17);
- (iv) a moving charge (current) *or a changing electric field* creates a magnetic field.

Item (iv) is the piece Ampère originally missed and Maxwell fixed. Its consequence is spectacular: take a region with no charges and no matter. A changing B makes an E , a changing E makes a B , and the pair detaches from whatever created it and travels outward as an *electromagnetic wave* at

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3.0 \times 10^8 \text{ m/s},$$

a number built purely from measured constants of electricity and magnetism. The waves carry energy; visible light, radio, and X-rays are all the same phenomenon at different wavelengths—which is exactly where FY7's optics section picks up.

Remark (How far the code analogy goes). Treating $\vec{E}$ and $\vec{B}$ as functions evaluated at sample points, and computing total fields by summing per-charge contributions, is precisely how a physics engine or a field solver works—superposition is linearity, so memoising and adding contributions is legitimate. The analogy breaks in two places. A field is not mere bookkeeping: it carries energy and momentum of its own (theorem 1.9 already stores W in the field, not in the plates), and deleting a charge does not update the rest of the universe instantly—the disturbance travels at c . There is also a floor to the classical field picture entirely: at low enough energies the field is quantised into photons, and no continuous function describes a single one of them.

Remark (Most common mistakes). a) Confusing force with field and energy with potential: keep track of the per-unit-charge quantities $(\vec{E}, V)$ versus the whole-charge ones $(\vec{F}, W_{\text{pot}})$.

- b) Forgetting that $\vec{F} = q\vec{v} \times \vec{B}$ needs $\sin \alpha$: a charge moving parallel to $\vec{B}$ feels nothing.
- c) Applying $E = kq/r^2$ or $U = kq/r$ at $r = 0$ —point charges are an idealisation that breaks at the origin.
- d) Dropping the minus sign of Lenz's law and getting the current direction backwards. Say the sentence “the induced current opposes the change” every single time.
- e) Mixing up Q and U as the fixed quantity in capacitor questions: with the battery connected, U is fixed; disconnected, Q is fixed.
- f) Using $r = mv/(qB)$ without checking that $\vec{v} \perp \vec{B}$; at an angle the path is a helix and only the perpendicular component of v sets the radius.

2 Exercises

Exercise 2.1. Charges $q_1 = +2.0 \mu\text{C}$ at $x = 0$ and $q_2 = -3.0 \mu\text{C}$ at $x = 40 \text{ cm}$ are fixed on the x -axis. A third charge $q_3 = +5.0 \mu\text{C}$ is placed at $x = 15 \text{ cm}$.

- (a) Find the force on q_3 due to q_1 and the force on q_3 due to q_2 : magnitude and direction of each.
- (b) Find the net force on q_3 (magnitude and direction).

- (c) At what x between the two fixed charges does the electric field vanish?

Exercise 2.2. Two large parallel plates are 3.0 mm apart and connected to a 400 V supply, so the field between them is uniform. An electron is released from rest at the negative plate.

- (a) Find the electric field strength between the plates.
- (b) Find the force on the electron and its acceleration.
- (c) Find the electron's speed when it reaches the positive plate, using energy.

Exercise 2.3. A parallel-plate capacitor has plate area 50 cm^2 , separation 0.50 mm, and a dielectric with $\epsilon_r = 4.0$ between the plates.

- (a) Find its capacitance.
- (b) Find the stored charge and energy when it is connected to 24 V.
- (c) The battery is disconnected and the plates are then pulled to twice their original separation. State what happens to C , U , and the stored energy, and explain where the extra work goes.

Exercise 2.4. A 12 V battery drives $R_1 = 150 \Omega$ in series with the parallel combination of $R_2 = 220 \Omega$ and $R_3 = 330 \Omega$.

- (a) Find the total resistance seen by the battery and the current it delivers.
- (b) Find the current through each resistor and the voltage across each.
- (c) Find the power dissipated in R_1 and the total power delivered by the battery.

Exercise 2.5. A proton enters a uniform magnetic field of $B = 0.25 \text{ T}$ at $v = 4.0 \times 10^5 \text{ m/s}$, perpendicular to the field.

- (a) Determine the direction of the initial force (describe it in words, choosing axes yourself).
- (b) Find the radius and the period of the resulting circular path.
- (c) An electron enters with the same velocity in the same field. How do its radius and period compare with the proton's? Justify from the formulas.

Exercise 2.6. Prove that a magnetic field does no work on a moving charge. Start from $\vec{F} = q\vec{v} \times \vec{B}$ and the definition of mechanical work, and deduce that a particle moving in a pure magnetic field keeps constant speed and constant kinetic energy.

Exercise 2.7. A metal rod of length $L = 50 \text{ cm}$ slides without friction along horizontal rails in a uniform vertical field $B = 0.20 \text{ T}$, at constant speed $v = 4.0 \text{ m/s}$. The rails are connected to a resistor $R = 5.0 \Omega$; the rod and rails have negligible resistance.

- (a) Starting from the magnetic force on the charge carriers in the rod, derive that the induced voltage across the moving rod is $U_{\text{ind}} = BLv$, and state the condition under which this formula holds.
- (b) Find the induced voltage, the induced current, and its direction (explain your reasoning with Lenz's law).
- (c) Find the external force needed to keep the rod at constant speed, and the mechanical power supplied. Compare with the power dissipated in the resistor.

Exercise 2.8. A proton moves in a uniform magnetic field $\vec{B} = (0, 0, B)$ in vacuum, with initial velocity in the xy -plane. Its motion obeys

$$\frac{d\vec{v}}{dt} = \frac{q}{m} \vec{v} \times \vec{B}.$$

Simulate this numerically in a language of your choice.

- (a) Implement the semi-implicit Euler step $\vec{v}_{n+1} = \vec{v}_n + \frac{q}{m} \Delta t (\vec{v}_n \times \vec{B})$, then $\vec{r}_{n+1} = \vec{r}_n + \Delta t \vec{v}_{n+1}$, and also plain (explicit) Euler where $\vec{r}$ is updated with $\vec{v}_n$.
- (b) Run both for a few periods with $\Delta t = T/100$ (estimate T from theorem 1.15) and plot the orbits. Measure how the particle's speed drifts over one period for each scheme.
- (c) Compare the simulated orbit radius with the theoretical $r = mv/(qB)$ and explain any discrepancy. State which scheme you would use in a real physics engine and why.

3 Solutions

Solution to exercise 2.1. (a) Using Coulomb's law, we have

$$|F| = k \frac{|q_1 q_2|}{r^2}$$

The force on q_3 due to q_1 :

$$|F_{31}| = k \frac{10.0 \cdot 10^{-12} C}{0.15m} \approx 0.6N$$

The direction is away from q_1 , since they are both positively charged. The force on q_3 due to q_2 :

$$|F_{32}| = k \frac{15.0 \cdot 10^{-12} C}{0.25cm} \approx 0.54N$$

This time the force points towards q_2 since they are oppositely charged.

- (b) The net force is just the sum of the previous 2 forces in this case.

$$\vec{F}_{net} = \vec{F}_{31} + \vec{F}_{32}$$

It is clear that $\vec{F}_{32}$ and $\vec{F}_{31}$ both point in the positive x-axis direction. Thus:

$$\begin{aligned} F_{net} &= F_{31} + F_{32} \\ F_{net} &= 0.6N + 0.54N = 1.14N \end{aligned}$$

- (c) The electric field is given by:

$$E = \frac{F_{net}}{q_0}$$

We have the following equation:

$$\begin{aligned}
E &= 0 \\
\frac{F_{net}}{q_0} &= 0 \\
\frac{\frac{q_1 q_0}{x^2} + \frac{q_2 q_0}{(40-x)^2}}{q_0} &= 0 \\
\frac{q_1}{x^2} + \frac{q_2}{(40-x)^2} &= 0 \\
\frac{(40-x)^2 q_1}{x^2} + q_2 &= 0 \\
\frac{(40-x)^2}{x^2} &= -\frac{q_2}{q_1} \\
\frac{40-x}{x} &= \sqrt{-\frac{q_2}{q_1}} \\
40-x &= x \sqrt{-\frac{q_2}{q_1}} \\
40 &= x \left(1 + \sqrt{-\frac{q_2}{q_1}}\right) \\
x &= \frac{40}{1 + \sqrt{-\frac{q_2}{q_1}}} \\
x &\approx 18cm
\end{aligned}$$

□

Solution to exercise 2.2. The electric field strength is given by:

$$E = \frac{U}{d} = 133\,333V/m$$

(b) The force is given by

$$F = qE$$

So the acceleration of the electron on is just:

$$\begin{aligned}
a &= \frac{eE}{m_e} \\
a &\approx 2.3 \cdot 10^{16} \frac{m}{s^2}
\end{aligned}$$

(c) The electric potential between the 2 plates is given by:

$$\Delta W_{pot} = q\Delta V$$

We also have:

$$\begin{aligned}
U_{AB} &= \frac{W_{AB}}{e} \\
400V \cdot e &= W_{AB} \\
400V \cdot e &= \frac{1}{2} m_e v^2 \\
v &= \sqrt{\frac{2 \cdot 400V \cdot e}{m_e}} \\
v &\approx 11.8 \cdot 10^{12} m/s
\end{aligned}$$

□

Solution to exercise 2.3. (a) Its capacitance is given by:

$$C = \varepsilon_0 \varepsilon_r \frac{A}{d}$$

We'll use $\varepsilon_0 = 8.85 \cdot 10^{-12} F/m$, $\varepsilon_r = 4.0 F/m$, $A = 0.005 m^2$ and $d = 0.005 m$. We have:

$$C \approx 3.5 \cdot 10^{-11} F$$

(b) We have:

$$Q = CU$$

So using the capacitance obtained earlier we have:

$$Q \approx 8.5 \cdot 10^{-10} C$$

(c) The battery is disconnected, let's treat Q as fixed and U as variable. The capacitance is half of what it used to be, since the separation is doubled. We have:

$$C_f = 1.75 \cdot 10^{-11} F = \frac{C}{2}$$

$$U_f = \frac{Q}{C_f} = 48V$$

The stored energy is given by:

$$W_f = \frac{1}{2} C_f U_f^2 \approx 2.03 \cdot 10^{-8} J$$

It is also clear that this energy is double of what it used to be. This is because separating the plates is doing work on the capacitor. $\square$

Solution to exercise 2.4. (a) The total resistance is given by:

$$\begin{aligned} R_t &= R_1 + R_{23} \\ R_t &= R_1 + \frac{R_2 R_3}{R_2 + R_3} \\ R_t &= 282 \Omega \end{aligned}$$

The current is given by:

$$\begin{aligned} I_{total} &= \frac{U_{total}}{R_{total}} \\ I_{total} &= \frac{12V}{282 \Omega} \approx 42 mA \end{aligned}$$

(b) For the whole circuit, we can think of the resistor pair R_{23} as one single resistor. Then, we see that R_1 and R_{23} are wired in series. We know that the voltage divides between resistors in series in proportion to their resistance. The voltage across R_1 is then:

$$\begin{aligned} U_1 &= U_{total} \frac{R_1}{R_t} \\ U_1 &= 12V \frac{150 \Omega}{282 \Omega} \\ U_1 &\approx 6.4V \end{aligned}$$

Then for the pair R_{23} , we have:

$$\begin{aligned} U_{23} &= U_{total} \frac{R_{23}}{R_t} \\ U_{23} &\approx 5.6V \end{aligned}$$

For the resistors R_2 and R_3 , the voltage stays the same ($5.6V$), but the current gets divided in proportion to the conductance ($G = \frac{1}{R}$):

$$I_{23} = I_{total} \quad (\text{Since } R_1 \text{ and } R_{23} \text{ are wired in series})$$

$$I_2 = I_{23} \frac{\frac{1}{R_2}}{\frac{1}{R_2} + \frac{1}{R_3}}$$

$$I_2 = I_{23} \frac{R_{23}}{R_2}$$

$$I_2 \approx 25mA$$

$$I_3 = I_{23} \frac{R_{23}}{R_3}$$

$$I_3 \approx 17mA$$

(c) The power dissipated in R_1 is given by:

$$P_1 = U_1 I_1$$

$$P_1 = 6.4V \cdot 42mA$$

$$P_1 = 0.27W$$

The total power given by the battery is given by:

$$P_{total} = U_{total} I_{total}$$

$$P_{total} = 12V \cdot 42mA$$

$$P_{total} = 0.5W$$

□

Solution to exercise 2.5. (a) Let's choose that the field points to the left (negative x), and that the proton is going forward (positive y). The z-axis points up. Using the right hand rule (middle finger represents the field, index finger represents the velocity and the thumb represents the resulting force), we deduce that the initial force vector points upwards, towards the z-axis. (b) The magnitude of the force is given by:

$$|F| = qvB \quad (\text{Since the velocity is perpendicular to the field})$$

Since the proton is perpendicular to the field, F is a centripetal force and the proton traces a circular path. Following the equations for circular motion:

$$a = \frac{v^2}{r}$$

$$F = \frac{mv^2}{r}$$

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv^2}{qvB}$$

$$r = \frac{mv}{qB}$$

$$r \approx 0.013m = 13mm$$

The period is just the circumference divided by the velocity:

$$T = \frac{2\pi r}{v} \approx 2.6 \cdot 10^{-7}s$$

(c) For an electron, the equations don't change, except that it has an equal and opposite charge compared to a proton. Then, the force will point in the opposite direction (negative z) and thus the circular motion will be the same but in the opposite direction. $\square$

Solution to exercise 2.6. Knowing that $\vec{F} = q\vec{v} \times \vec{B}$ already tells us a crucial detail: the force exerted on a particle in a magnetic field is always perpendicular to the velocity. Let's see the definition of mechanical work:

$$\begin{aligned} W &= \int \vec{F} \cdot d\vec{s} \\ W &= \int \vec{F} \cdot \vec{v} dt \\ W &= \int (q\vec{v} \times \vec{B}) \cdot \vec{v} dt \\ W &= q \int (\vec{v} \times \vec{B}) \cdot \vec{v} dt \end{aligned}$$

The integrand is now a scalar triple product. A crucial identity of the scalar triple product is that $\vec{a} \cdot (\vec{a} \times \vec{b}) = 0$. Thus, the integral evaluates to 0 and the work done is also 0. From this we can deduce that the mechanical energy is conserved. Furthermore, since the force is always perpendicular to the velocity, we know that the magnitude of the velocity (speed) is also constant, and from this follows that the kinetic energy is also constant. $\square$

Solution to exercise 2.7. Each charge carrier experiences a force from the magnetic field given by:

$$\vec{F} = q\vec{v} \times \vec{B}$$

These forces apply to each electron so the charges build up to one end of the rod. They will then achieve an equilibrium once the electric field they build up cancels out the aforementioned force, so:

$$\begin{aligned} qE &= q\vec{v} \times \vec{B} \\ qE &= qvB \quad (\text{Since the magnetic field is perpendicular to the velocity}) \\ E &= vB \end{aligned}$$

To get the tension, we just need the difference in potential, which scales linearly in proportion to the length:

$$\begin{aligned} U_{ind} &= LE \\ U_{ind} &= BLv \end{aligned}$$

There's also a more simple explanation for the formula. If we take the whole circuit as the coil, then its flux changes in proportion to its area. The area grows Lv each second, so the flux in this scenario grows BLv each second which is exactly the induced voltage. This formula holds when the magnetic field is perpendicular to the circuit and the velocity of the rod. Otherwise we'd have to take cross products for the velocity and dot products for the coil.

(b) Using the aforementioned formula, the induced voltage is:

$$\begin{aligned} U_{ind} &= BLv \\ U_{ind} &= 0.2T \cdot 0.5m \cdot 4.0m/s \\ U_{ind} &= 0.4V \end{aligned}$$

The induced current can be calculated using the relation $I = \frac{U}{R}$:

$$I_{ind} = \frac{0.4V}{5.0\Omega} = 80mA$$

Since the uniform magnetic field points up, the induced magnetic field must point down. Using the right hand rule (curling the fingers in the induced magnetic field gives us the direction of

current), we can deduce that the current must flow counter-clockwise. (The electrons flow in the opposite direction).

(c) Since the magnetic field doesn't do any work, we can assert that energy is conserved in the system:

$$\begin{aligned}
E_k + E_e &= E_0 \\
\frac{1}{2}mv(t)^2 + Pt &= \frac{1}{2}mv_0^2 \\
\frac{1}{2}mv(t)^2 + UIt &= \frac{1}{2}mv_0^2 \\
\frac{1}{2}mv(t)^2 + \frac{U^2}{R}t &= \frac{1}{2}mv_0^2 \\
\frac{1}{2}mv(t)^2 + \frac{(BLv(t))^2}{R}t &= \frac{1}{2}mv_0^2 \\
v(t)^2\left(\frac{1}{2}m + \frac{(BL)^2t}{R}\right) &= \frac{1}{2}mv_0^2 \\
v(t)^2 &= \frac{\frac{1}{2}mv_0^2}{\frac{1}{2}m + \frac{(BL)^2t}{R}} \\
v(t)^2 &= \frac{mv_0^2}{m + \frac{2(BL)^2t}{R}} \\
v(t)^2 &= \frac{Rmv_0^2}{Rm + 2(BL)^2t} \\
v(t)^2 &= \frac{Rmv_0^2}{Rm + 2B^2L^2t} \\
v(t) &= \sqrt{\frac{Rmv_0^2}{Rm + 2B^2L^2t}}
\end{aligned}$$

We observe that $v(t)$ is decreasing. To keep the velocity constant, we need to do work that is equal to the energy dissipated in the resistor.

$$\begin{aligned}
W &= E_e \\
W &= UIt \\
W &= \frac{U^2t}{R} \\
W &= \frac{(BLv)^2t}{R}
\end{aligned}$$

Let's find the external force from the work:

$$\begin{aligned}
W &= Fs \\
Fs &= \frac{(BLv)^2t}{R} \\
Fvt &= \frac{(BLv)^2t}{R} \\
Fv &= \frac{(BLv)^2}{R} \\
F &= \frac{(BL)^2v}{R}
\end{aligned}$$

Since the velocity stays constant in this scenario, we can calculate F:

$$F = \frac{(0.2T \cdot 0.5m)^2 4.0m/s}{5.0\Omega} = 0.008N$$

The power dissipated in the resistor is simply.

$$P = UI$$

$$P = \frac{U^2}{R}$$

$$P = \frac{(BLv)^2}{R}$$

$$P = \frac{(0.2T \cdot 0.5m \cdot 4.0m/s)^2}{5.0\Omega}$$

$$P = 0.032W = Fv$$

□

4 Feedback